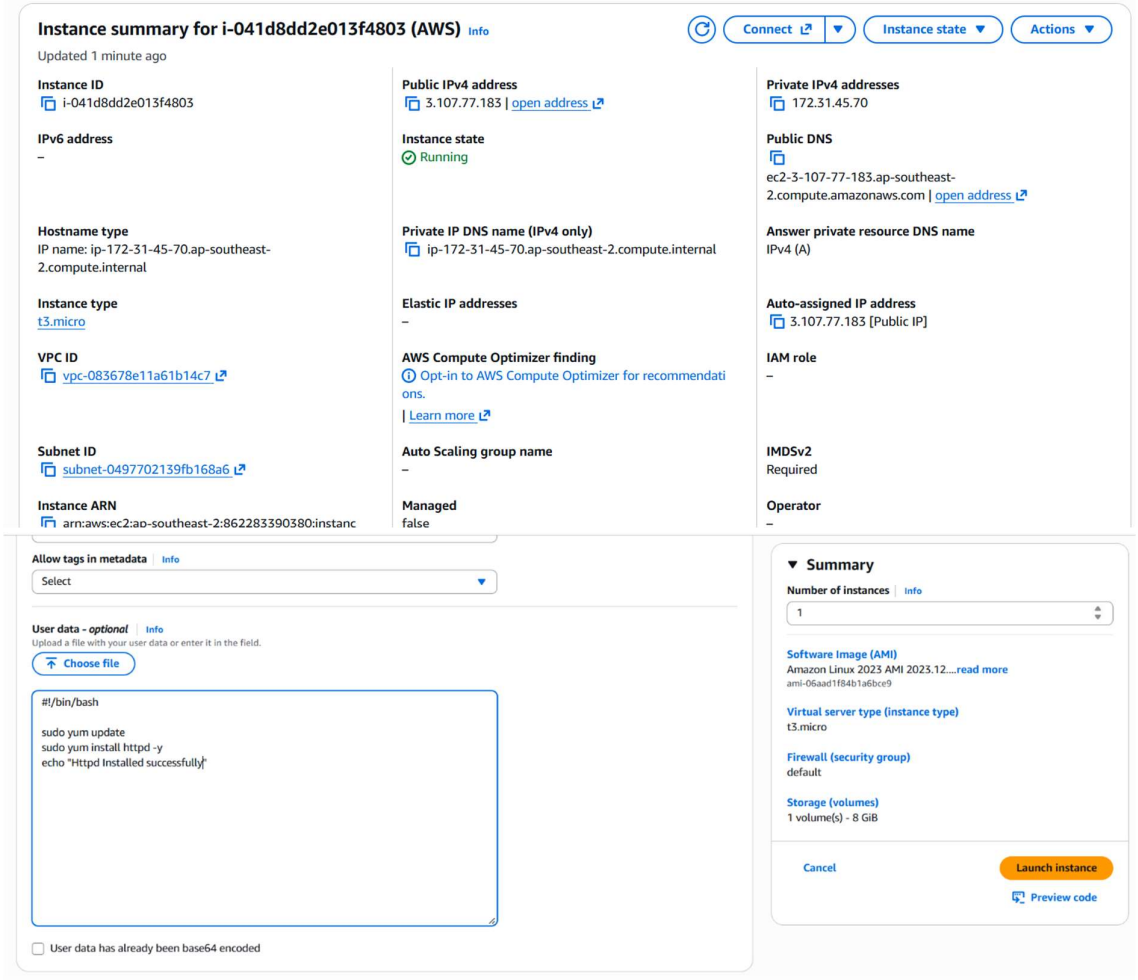
AWS Task

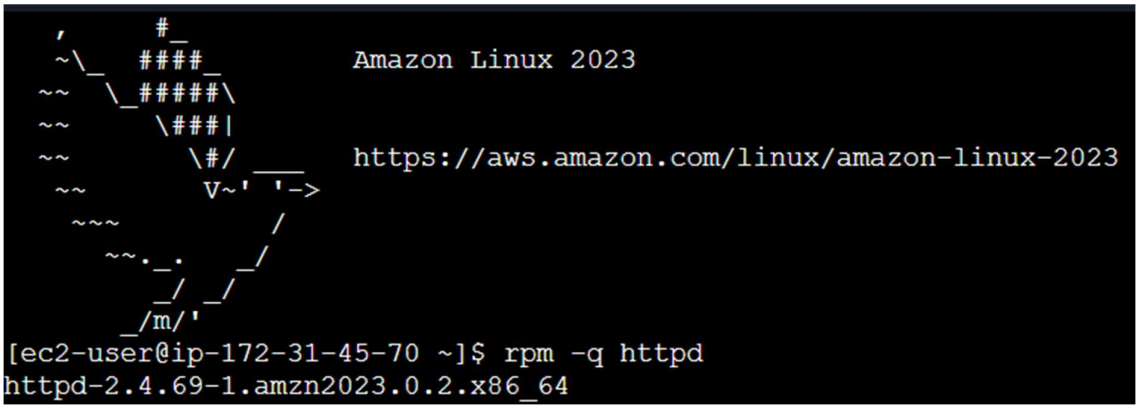
1. Launch one EC2 using Amazon Linux 2 image and add a script in user data to install Apache.

Step 1:- Create a instance. While creating select amazon linux image and then go to advanced option scroll down and we can see user data.

Step 2:- Write the script to install the apache

Step 3:- Then after connecting the instance we can check whether the package is installed or not. We can use rpm or directly view in etc folder



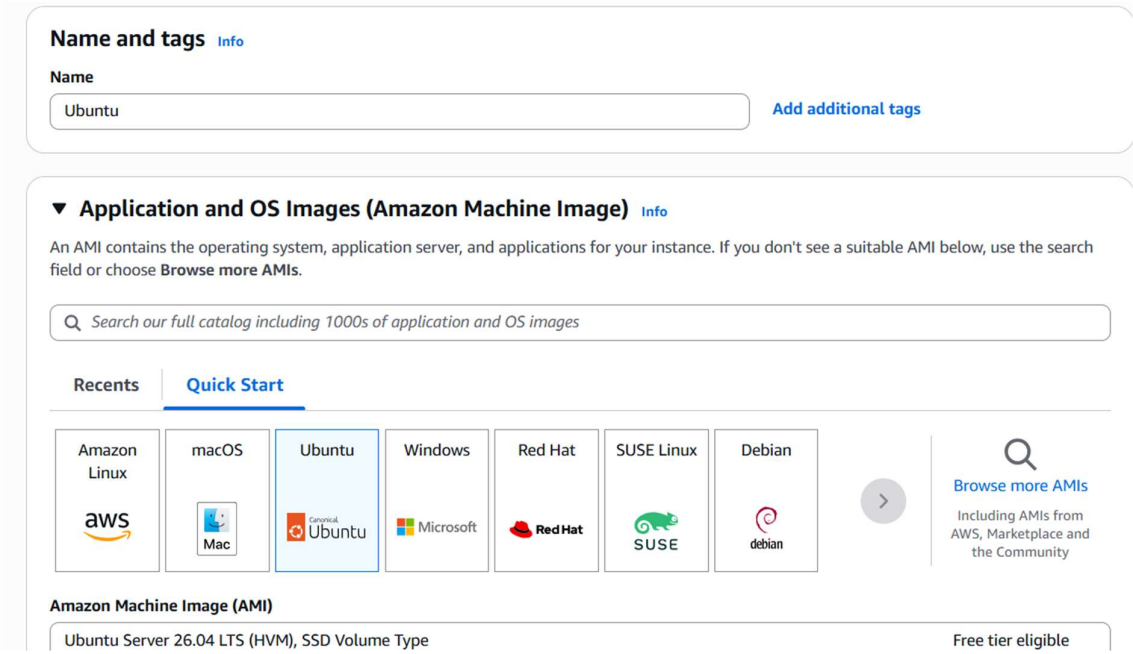


2. Launch one EC2 using Ubuntu image and add a script in user data to install Nginx.

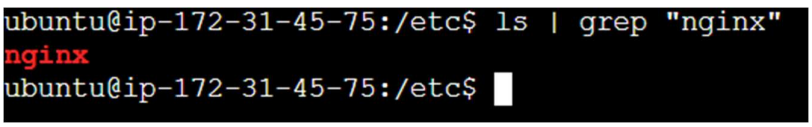
Step 1:- Create a instance. While creating select Ubuntu image and then go to advanced option scroll down and we can see user data.

Step 2:- Write the script to install the nginx.

Step 3:- Then after connecting the instance we can check whether the package is installed or not. We can directly view in etc folder if package is intsalled or not







3. Launch one Windows server and install Tomcat on Windows.

Step 1:- Create a instance with window image. Configure the RDP port in the security group

Step 2:- Then using Remote desktop connection connect to the instance.

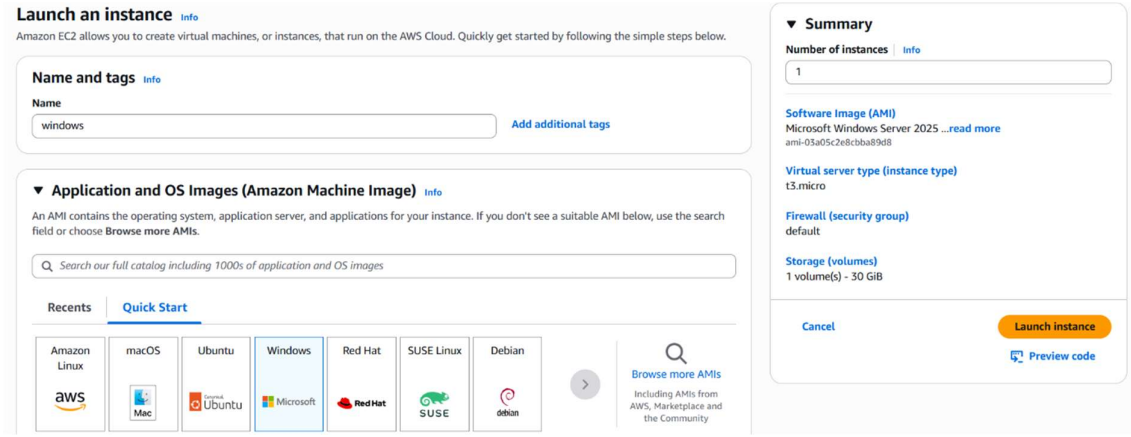
Step 3:- To install tomcat we need Java installed into the windows

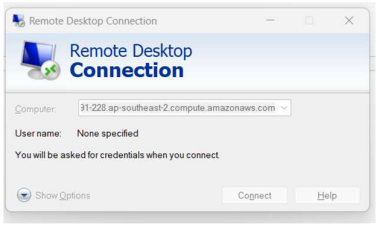
Step 4:- use command prompt and winget to download openjdk

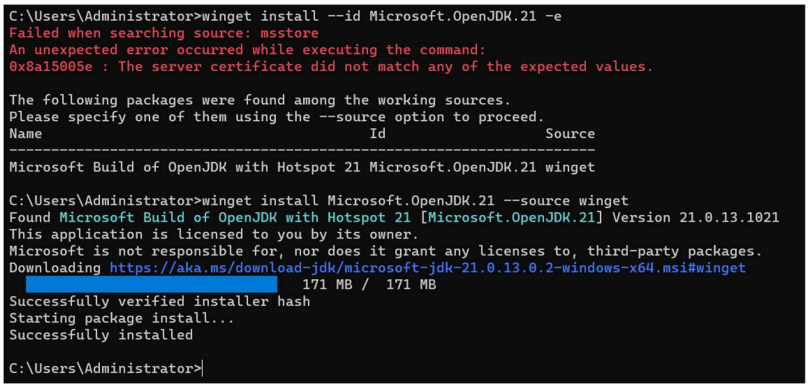
Step 5:- Go to web browser and download tomcat installer.

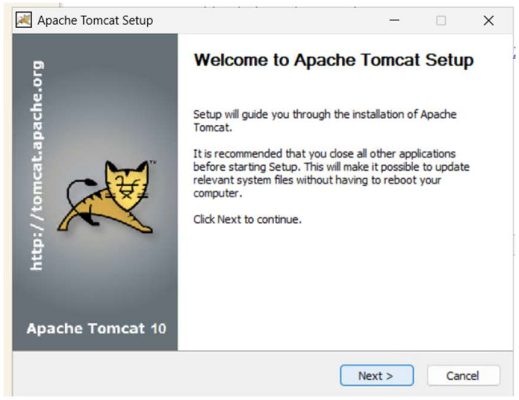
Step 6:- Configure OpenJDK path to tomcat.

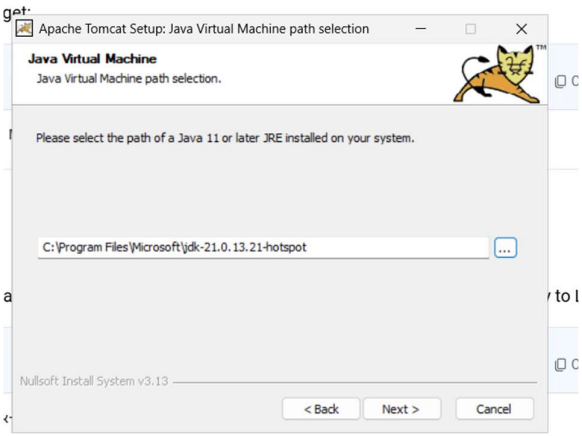
Step 7:- After finishing installation we can open localhost:8080 to check whether it is running or not.

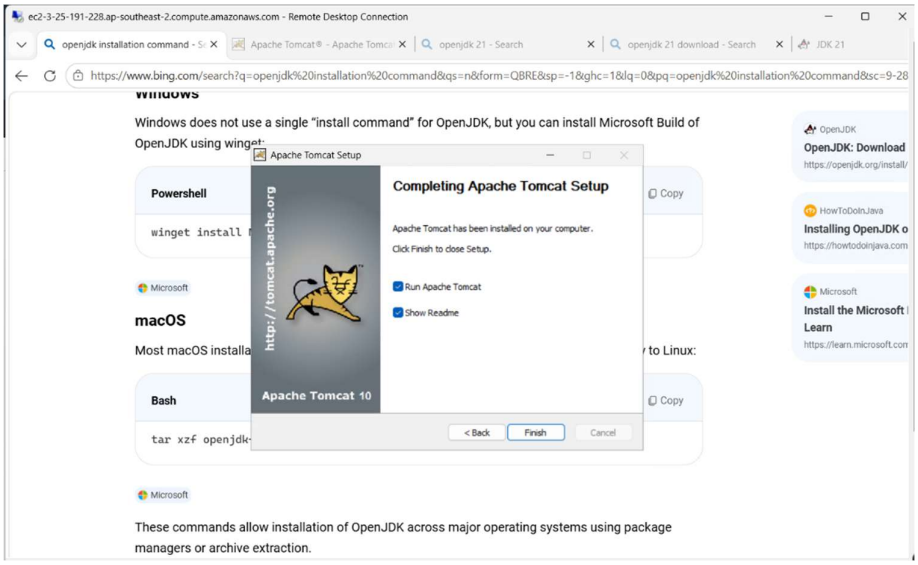


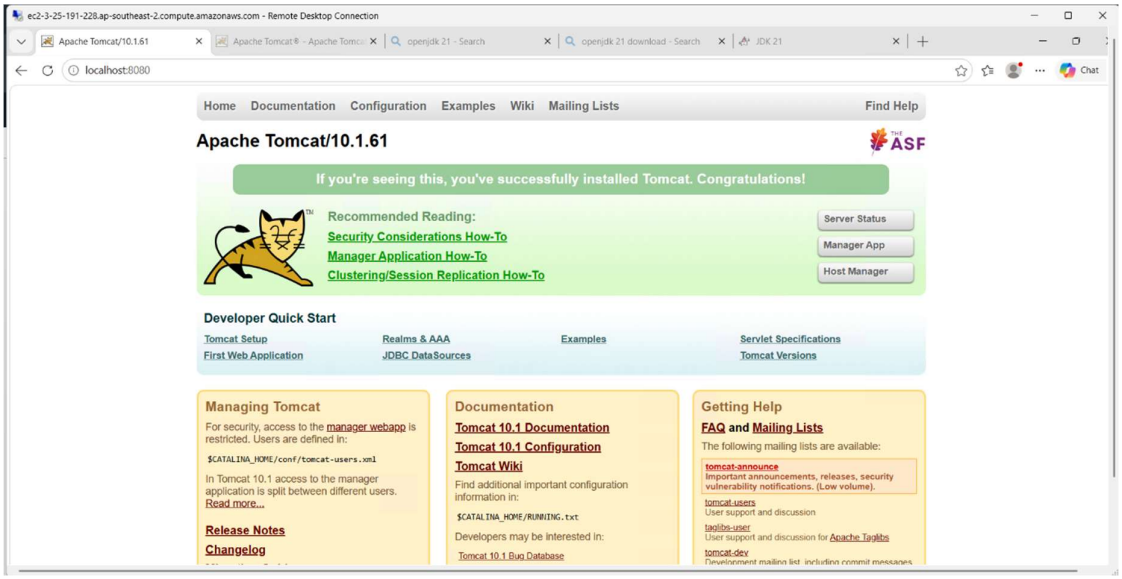










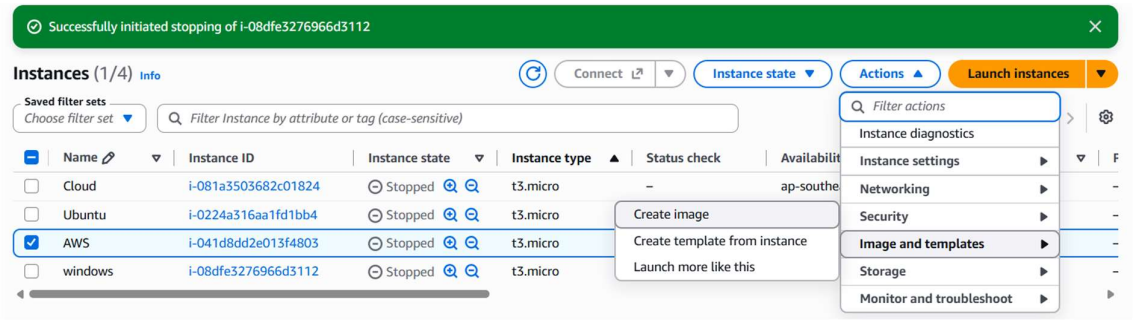


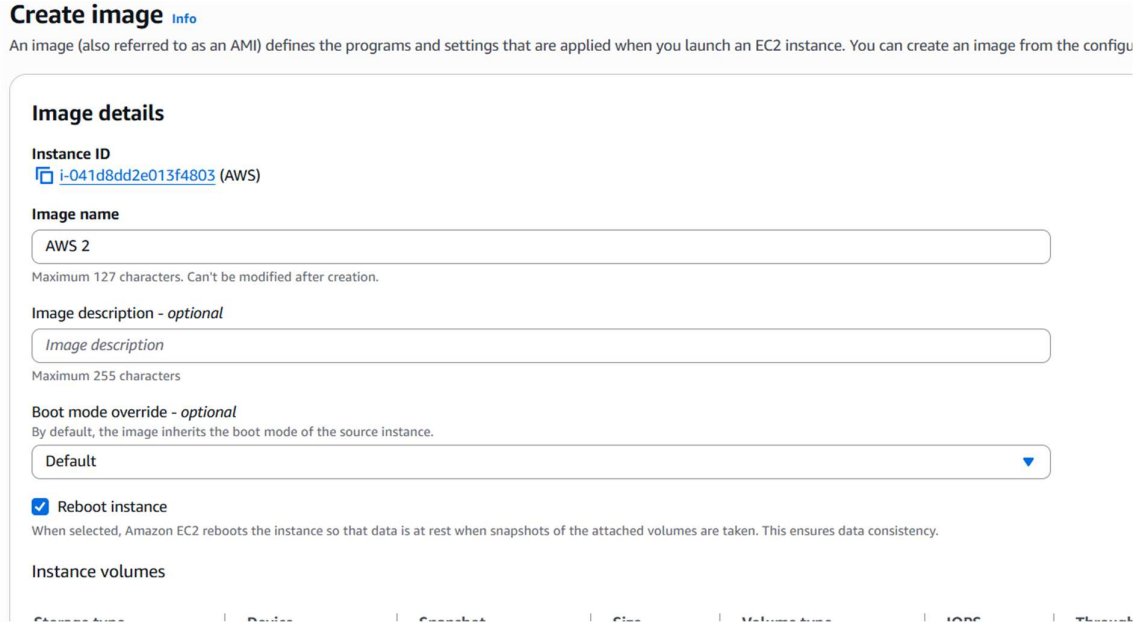
4. Take a snapshot of the instance created in Task 1.

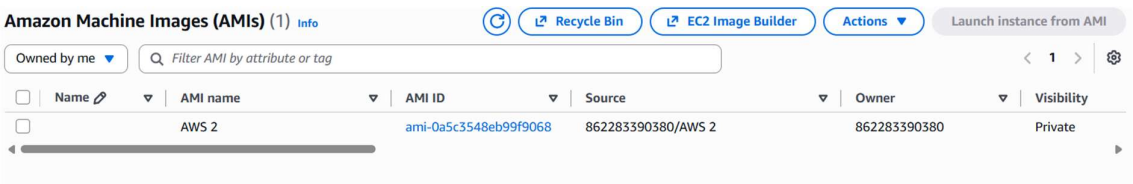
Step 1:- Select the instance go to actions and image

Step 2:- Name the image then scroll down then create image

Step 3:- Now go to AMI to check whether the image is present or not





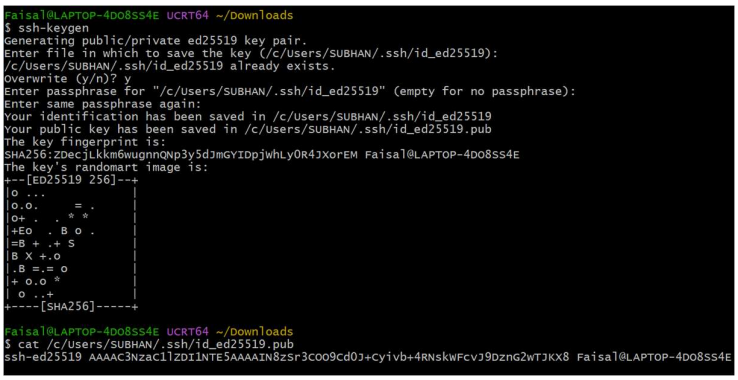


5. Assign passwordless authentication for the EC2 created in Task 2.

Step 1:- First generate a ssh-keygen. Then copy it and paste it into authorized_keys in /root/.ssh/ path

Step 2:- Now go to sshd_config and change PasswordAuthentication from no to yes. Its should not be commented

Step 3:- Now try to connect the instance with pem key



```
root@ip-172-31-45-75:/home/ubuntu# cd .ssh
root@ip-172-31-45-75:/home/ubuntu/.ssh# ls
authorized_keys
root@ip-172-31-45-75:/home/ubuntu/.ssh# vi authorized_keys
root@ip-172-31-45-75:/home/ubuntu/.ssh# cat authorized_keys
ssh-ed25519 AAAAC3NzaC11ZDI1NTE5AAAAIN8zSr3COO9Cd0J+Cyivb+4RNskWFcvJ9DznG2wTJKX8 Faisal@LAPTOP-4DO8SS4E

ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQDydJ1Mqt+pVmpCLYDIyV51ESL5rP/zZT55b1u1psx4oNH7YVOTbw3Q3E1bVUxp6jz56MNIsR69HSLlza6d0
hexzCVf4X4DRWDXQPTQ7+WTAVroGRarT3TiFBWCotXVgaS71tqBpcEh/U+p8TGr0kT+FJRl6RG8Vs4loAifcltUk6sj/zODAjLqGSe43vmPMl2ZL+DN7i+T15
jEfLOjnPAeZS7UNLkWHiudAl1eYKx/j36NzQGW9Urf jenkins
root@ip-172-31-45-75:/home/ubuntu/.ssh#
```

```
# To disable tunneled clear text passwords, change to "no" here!
PasswordAuthentication yes
#PermitEmptyPasswords no
```

```
$ ssh ubuntu@3.24.216.37
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-1014-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Oct 29 11:12:50 UTC 2026

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   33.9% of 6.61GB   Processes:             125
  Memory usage: 31%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.45.75


Expanded Security Maintenance for Applications is not enabled.

38 updates can be applied immediately.
10 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

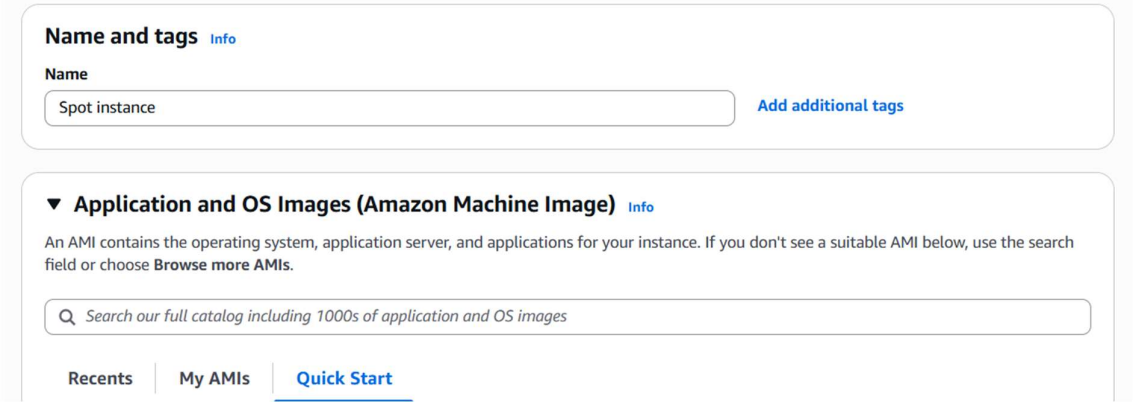

Last login: Thu Oct 29 10:31:11 2026 from 13.239.158.4
ubuntu@ip-172-31-45-75:~$
```

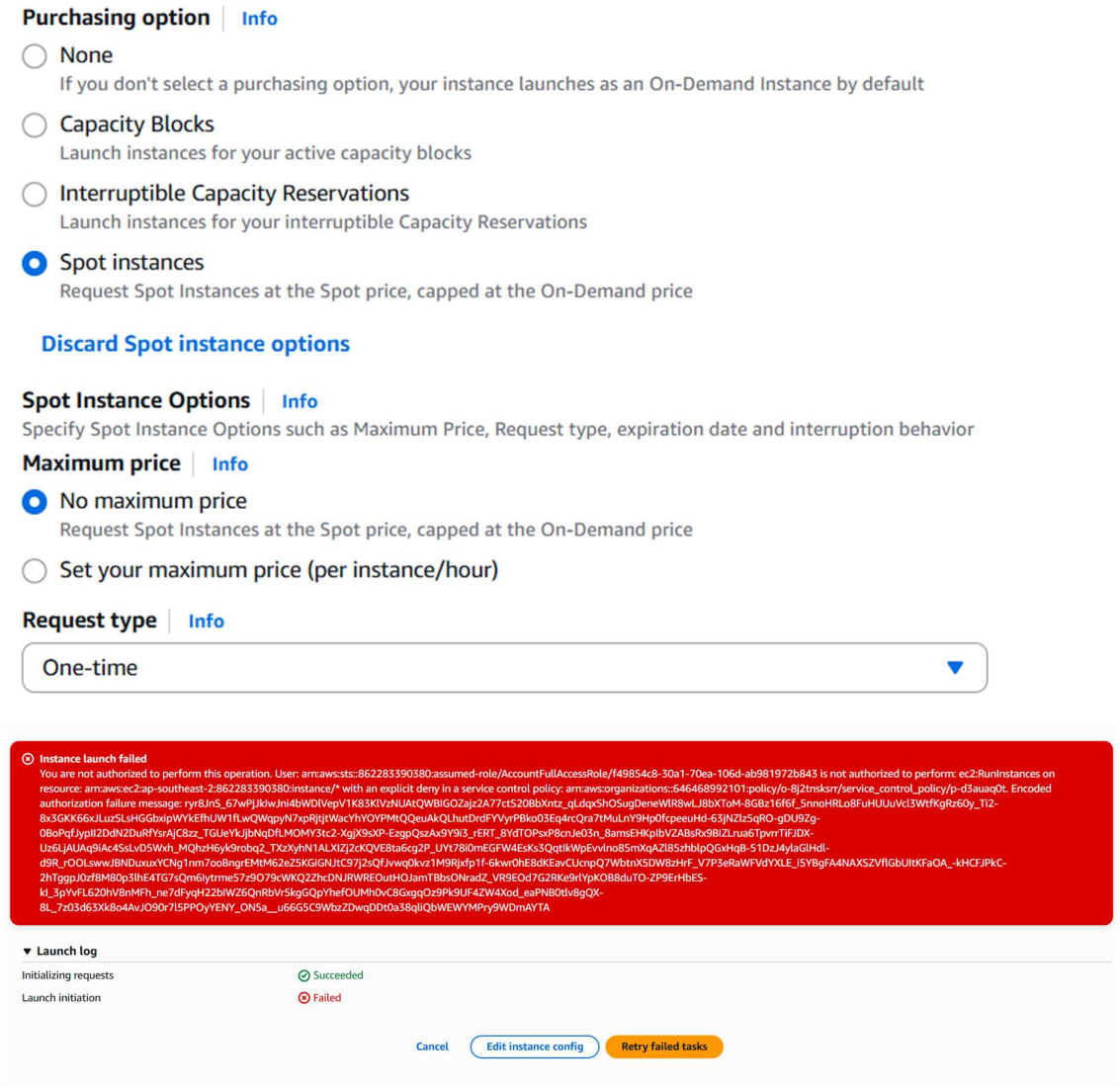
6. Launch any EC2 using the spot purchasing option.

Step 1:- Launch a new instance configure key pair, security groups.

Step 2:- Open Advanced Settings and scroll down to purchase option select spot instance.

Step 3:- Then launch Instance.





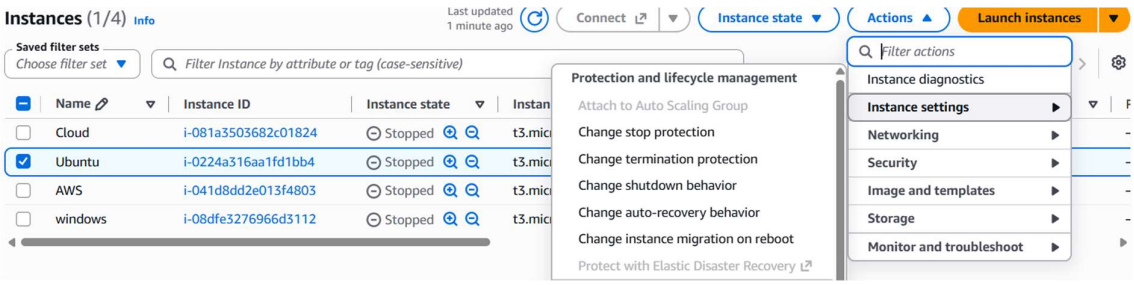
7. Enable termination policy on the EC2 created in Task 2.

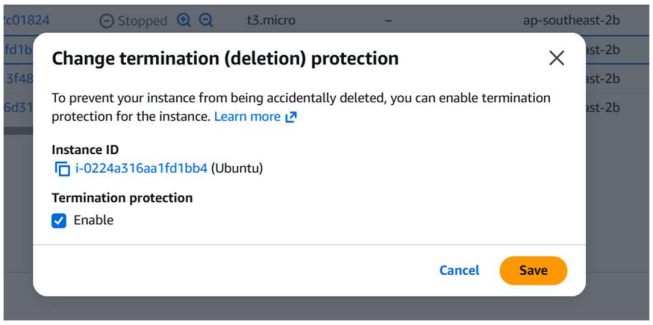
Step 1:- Select the instance and then go to action.

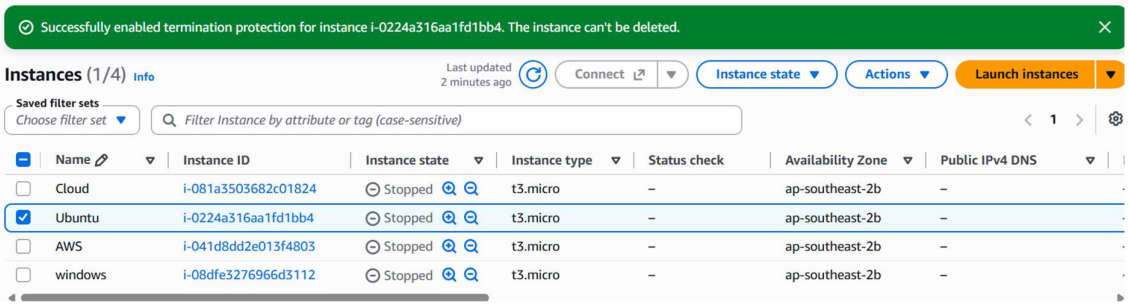
Step 2:- In actions go to instance settings and then to change termination protection.

Step 3:- Enable the policy then save.

Step 4:- Now it is successfully enabled







8. Launch one EC2 using AWS CLI.

Step 1:- Create a IAM user with amazonec2fullaccess and admin access

Step 2:-Then click on create access key to create public access key and secret access key.

Step 3:- Open cmd check whether AWS cli is installed or not. If not install it

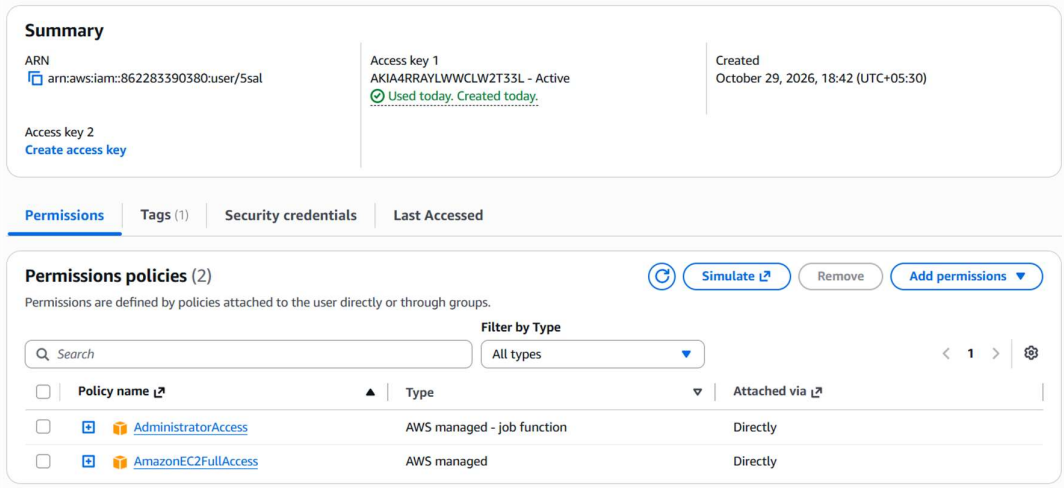
Step 4:- First configure access key, secret key and availability zone we are using, also output format to json.

Stop 5:- Fetch the AMI image and configure it into run instance command

Stop 6:- Also configure the Pem key file we can select any file.

Stop 7:- We have ssm-parameter and describe image parameter to get AMI id.

Step 8:- After using run instance command instance is created.



```
C:\Windows\System32>aws configure
AWS Access Key ID [****************FHL2]: AKIA4RRAYLWWCLW2T33L
AWS Secret Access Key [****************8TP5]: UgqWUyP5b/LKWbNHlQe6xJ5BMpkuGGzzGTrAsT2b
Default region name [ap-south-1]:
Default output format [json]: json

Configure AWS skills and the AWS MCP server for your AI coding agent(s)? [y/n/never]: never

C:\Windows\System32>aws sts get-caller-identity
{
    "UserId": "AIDA4RRAYLWWG5U7AQIZ5",
    "Account": "862283390380",
    "Arn": "arn:aws:iam::862283390380:user/5sal"
}
```

```
C:\Windows\System32>aws ssm get-parameter --name /aws/service/ami-amazon-linux-latest/al2023-ami-kernel-default-x86_64 --region ap-south-1 --query "Parameter.Value" --output text

aws: [ERROR]: An error occurred (AccessDeniedException) when calling the GetParameter operation: User: arn:aws:iam::862283390380:user/5sal is not authorized to perform: ssm:GetParameter on resource: arn:aws:ssm:ap-south-1::parameter/aws/service/ami-amazon-linux-latest/al2023-ami-kernel-default-x86_64 with an explicit deny in a service control policy: arn:aws:organizations::646468992101:policy/o-8j2tnsksrr/service_control_policy/p-4wwxq65z

C:\Windows\System32>aws ssm get-parameter --name /aws/service/ami-amazon-linux-latest/al2023-ami-kernel-default-x86_64 --region ap-south-1

aws: [ERROR]: An error occurred (AccessDeniedException) when calling the GetParameter operation: User: arn:aws:iam::862283390380:user/5sal is not authorized to perform: ssm:GetParameter on resource: arn:aws:ssm:ap-south-1::parameter/aws/service/ami-amazon-linux-latest/al2023-ami-kernel-default-x86_64 with an explicit deny in a service control policy: arn:aws:organizations::646468992101:policy/o-8j2tnsksrr/service_control_policy/p-4wwxq65z

C:\Windows\System32>aws configure set region ap-southeast-2

C:\Windows\System32>aws configure get region
ap-southeast-2

C:\Windows\System32>aws ssm get-parameter --name /aws/service/ami-amazon-linux-latest/al2023-ami-kernel-default-x86_64 --region ap-southeast-2 --query "Parameter.Value" --output text
ami-06aad1f84b1a6bce9


C:\Windows\System32>aws ec2 run-instances --image-id ami-06aad1f84b1a6bce9_--instance-type t3.micro --key-name jenkins --count 1 --region ap-southeast-2 --tag-specifications "ResourceType=instance,Tags=[{Key=Name,Value=Faisal-EC2}]"
```


```
{
    "ReservationId": "r-0c4c81d8bf172d2fd",
    "OwnerId": "862283390380",
    "Groups": [],
    "Instances": [
        {
            "Architecture": "x86_64",
            "BlockDeviceMappings": [],
            "ClientToken": "dd999c44-cccf-4309-8ca9-0f112bc867f3",
            "EbsOptimized": false,
            "EnaSupport": true,
            "Hypervisor": "xen",
            "NetworkInterfaces": [
                {
                    "Attachment": {
                        "AttachTime": "2026-10-29T13:23:23+00:00",
                        "AttachmentId": "eni-attach-0515eed59b1a7ec11",
                        "DeleteOnTermination": true,
                        "DeviceIndex": 0,
                        "Status": "attaching",
                        "NetworkCardIndex": 0
                    },
                    "Description": "",
                    "Groups": [
                        {
                            "GroupId": "sg-0632a39310deef8d2",
                            "GroupName": "default"
                        }
                    ],
                    "Ipv6Addresses": [],
                    "MacAddress": "06:ff:c2:93:50:71",
                    "NetworkInterfaceId": "eni-099c754431004c0c9",
                    "OwnerId": "862283390380",
                    "PrivateDnsName": "ip-172-31-44-81.ap-southeast-2.compute.internal",
                    "PrivateIpAddress": "172.31.44.81",
                    "PrivateIpAddresses": [
                        {
                            "Primary": true,
                            "PrivateDnsName": "ip-172-31-44-81.ap-southeast-2.compute.internal",
                            "PrivateIpAddress": "172.31.44.81"
                        }
                    ],
                    "SourceDestCheck": true,
                    "Status": "in-use",
                    "SubnetId": "subnet-0497702139fb168a6",
                    "VpcId": "vpc-083678e11a61b14c7",
                    "InterfaceType": "interface",
                    "Operator": {
                        "Managed": false
                    }
                }
-- More  --
```

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	F
Cloud	i-081a3503682c01824	Stopped	t3.micro	–	ap-southeast-2b	–	–
Ubuntu	i-0224a316aa1fd1bb4	Stopped	t3.micro	–	ap-southeast-2b	–	–
AWS	i-041d8dd2e013f4803	Stopped	t3.micro	–	ap-southeast-2b	–	–
windows	i-08dfe3276966d3112	Stopped	t3.micro	–	ap-southeast-2b	–	–
Faisal-EC2	i-09eecf027bce6cd15	Running	t3.micro	Initializing	ap-southeast-2b	ec2-52-65-132-234.ap-...	5